No. of Printed Pages : 4
Roll No. ....

220643

4th Sem / Chemical (Pulp & Paper)

Subject : Heat Transfer

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Heat Exchanger uses

a) One fluid only b) Two fluids

c) Three Fluids d) Solid Only

Q.2 Heat transfer in straight line without any medium is

a) Conduction b) Convection

c) Radiation d) All

Q.3 Unit of Diffusivity

a) m^2/hr b) m^2/K

c) hr/m^2 d) None

Q.4 Which material is best conductor of heat?

a) Copper b) Wood

c) Glass d) Air

Q.5 Nusselt number is used in

a) Radiation b) Conduction

c) Convection d) All

Q.6 Emissivity of a perfect black body is

a) 0 b) 0.5

c) 1 d) Infinity

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define black body in Radiation.

Q.8 Write the Planck's law.

Q.9 What is baffle.

Q.10 Full form of LMTD.

Q.11 Fourier's law equation________.

Q.12 Unit of Thermal conductivity is________

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Derive One-dimensional heat conduction equation for a slab.

Q.14 Compare conduction, convection and Radiation.

Q.15 Explain the concept of Thermal Resistance.

Q.16 Explain Dittus Boelter's equation and its significance.

Q.17 Discuss about insulation of heat exchangers.

Q.18 Explain overall Heat Transfer Coefficient with its units.

Q.19 How scale formation occur in heat Exchangers. how we can prevent it?

Q.20 Explain Optimum Thickness of insulation.

Q.21 Describe Kirchoff's law in Radiation.

Q.22 A flat plate of area $1m^2$ at 800K radiates heat to a surrounding surface at 300K. If emissivity is 0.7 . Calculate the net radiative heat transfer rate.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the working of shell and Tube heat exchanger with diagram.

Q.24 Explain any two-feeding methods of evaporators with diagram.

Q.25 What is dimensional analysis . Write about Reynold's Number, Prondil number, Stanton number & packet number with their formula.